



ECP and BVA of TradXSell

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ECP Analysis

1. User Registration

Conditions	Valid Partitions	Invalid Partitions	Valid Boundaries	Invalid Boundaries
Username: Non-empty (1-50 letters)	- Alphabetic (1-50 letters) - Mixed case alphabetic (e.g., JohnDoe, johndoe)	- Empty string - Numeric (e.g., 12345) - Special characters (e.g., @user!) - Length > 50	- Length = 1 (e.g., J) - Length = 50 (e.g., J... 50 times)	- Length = 0 (e.g., "") - Length = 51 (e.g., J... 51 times)
Email: Valid format	- Proper structure (e.g., user@gmail.com) - Multiple subdomains (e.g., user@sub.gmail.com)	- Missing '@' (e.g., usergmail.com) - No domain (e.g., user@) - Invalid chars (e.g., user@@gmail.com)	- Minimal valid email (e.g., a@gmail.com)	- Missing domain (e.g., user@)
Password: Strong	- Minimum 8 chars, includes uppercase, lowercase, number, special character (e.g., Abc@1234)	- < 8 characters (e.g., Ab@123) - Missing any required element (e.g., no special char: Abc12345)	- Length = 8, meets all criteria (e.g., A1@bcdEF)	- Length = 7 (e.g., Abc@12)
	- More than 8 chars, valid (e.g., Strong!Pass123)	- Only uppercase (e.g., ABCDEFGH) - Only special chars (e.g., @@@@#@@) - Only lowercase (abcdefg)	- Longest reasonable valid password length (e.g., 50 chars)	- Length > 50
Phone Number: Numeric, valid Pakistani number	- Correct format: 03XXXXXXXXX - Valid prefixes: 030X, 031X, 034X (e.g., 03451234567)	- Invalid prefix (e.g., 0401234567) - Non-numeric (e.g., 0345ABCD567) - Less than 11 digits (e.g., 03451)	- Valid prefix, length = 11 (e.g., 03012345678)	- Length = 10 or less, or > 11 (e.g., 0301234)
	- Alternate valid prefix: 035X - Properly padded	- Empty input - Extra characters (e.g., 03451234567!)		Invalid format with special char

	numeric (e.g., 03400001234)			(e.g., 0345-1234567)
Address: Non-empty	- Simple address (e.g., 123 Main St) - Detailed address with special chars (e.g., House #10, Lane 4)	- Empty input - Non-meaningful input (e.g., @@@@)	- Single character (e.g., A)	- Empty string ("")
	- Minimal valid address (e.g., Apt 1) - Valid long address (e.g., Building X, Road Y, City Z)	- Only numeric with no context (e.g., 123456)		

2. User Login

Conditions	Valid Partitions	Invalid Partitions	Valid Boundaries	Invalid Boundaries
Email: Registered and valid format	- Registered email in the correct format (e.g., user@gmail.com)	- Unregistered email (e.g., newuser@gmail.com) - Invalid format (e.g., missing '@': usergmail.com)	- Minimal valid email (e.g., a@gmail.com)	- Missing '@' or domain (e.g., user@)
	- Variations of valid format (e.g., user+alias@gmail.com)	- Invalid domain (e.g., user@gmail) - Invalid characters (e.g., user@@gmail.com)	- Valid length boundary (e.g., 50 chars)	- Length > 50 (e.g., verylongemail..)
Password: Matches registered account	- Correct password associated with the email	- Incorrect password (e.g., mismatched with the registered email)	- Minimal length, matches registered (e.g., Pass@123)	- Minimal valid format but wrong (e.g., Wrong@1)
	- Strong password with valid length (e.g., Strong@1234)	- Passwords missing required elements (e.g., password,	- Longest valid password	- Too short (e.g., 5 chars: Ab@12)

		12345678, ALLUPPERCASE)	(e.g., 50 chars)	
		- Empty password	- Length = 8 with correct criteria	- Length = 7 or less
Account Status: Registered/Active	- Registered account in the system	- Unregistered email	- Active account	- Account locked or deactivated
		- Suspended/locked account		

3. Browse product by search

Conditions	Valid Partitions	Invalid Partitions	Valid Boundaries	Invalid Boundaries
Search Input: Product Name	- Exact product name (e.g., Laptop, Shirt)	- Non-existent product name (e.g., UnicornLaptop)	- Minimum valid input (e.g., 1 character: A)	- Empty input ("")
	- Partial product name (e.g., Lap, Shir)	- Special characters only (e.g., @\$%^)	- Maximum valid input length (e.g., 50 chars)	- Length > 50 (e.g., ProductName...)
	- Case-insensitive search (e.g., laptop, LAPTOP)	- Numeric input for text-only fields (e.g., 12345)	- Mixed alphanumeric valid input (e.g., Laptop123)	Invalid mixed input (e.g., Prod@123)
Search Results: Products available	- Products exist for the search term (e.g., Laptop returns relevant items)	- No products exist for the search term	- Minimal results (e.g., 1 product displayed)	- Zero results but search term valid
	- Products in multiple categories (e.g., Laptop found in electronics, accessories)	- Product exists but incorrectly categorized (e.g., Laptop in clothing)	- Maximum results displayable (e.g., 100 products)	Results with unrelated items included

Product Details: Correct display	- All details visible: name, price, description, size/color attributes	- Missing details (e.g., no description or price not displayed)	- Minimal valid detail set (name, price)	Missing key details (e.g., only name shown)
	- Dropdown options working for attributes like size and color	- Dropdown options missing, unclickable, or incorrect values displayed	- All dropdowns correctly pre-populated	Invalid or missing dropdown values
System Response: Navigation success	- Product details page loads successfully	- Error in redirection to product detail page (e.g., blank page, error message: Unable to display product details)	- Minimal valid redirection time (e.g., < 2 sec)	Slow or no redirection (> 5 sec)

4. Add Product to cart

Conditions	Valid Partitions	Invalid Partitions	Valid Boundaries	Invalid Boundaries
Dropdown Options: Selected properly	- All required dropdowns selected (e.g., color: Red, size: M)	- Missing required dropdowns (e.g., size not selected)	- Minimal valid selection (e.g., color: Red)	- No selection for one or more dropdowns
	- Optional dropdowns selected correctly (e.g., material: Cotton)	- Invalid values in dropdowns (e.g., non-existent option: Pink for size)	- Longest valid dropdown list handled correctly	Invalid option or empty dropdown values
Quantity: Positive integer	- Positive integer for quantity (e.g., 1, 5, 100)	- Zero or negative integer (e.g., 0, -1, -10)	- Minimum valid quantity (e.g., 1)	- Quantity = 0 or negative
	- Large positive integer (e.g., 9999)	- Non-integer input (e.g., decimal: 1.5, string: One, special chars: @)	- Largest reasonable valid quantity (e.g., 1000)	Quantity exceeding system limit (e.g., 10001)

Add to Cart Button: Clickable	- Button is active and clickable when all required conditions are met	- Button is inactive or unresponsive	- Click success with minimal conditions met	- Button unresponsive due to system error
System Validation: Success message	- Product added with a success message (e.g., "Product added to cart")	- Error message displayed due to invalid/missing input (e.g., "Please select all required options")	- Correct message shown upon valid addition	- Incorrect or no error message shown
Cart Status: Product added	- Product is reflected in the cart with correct attributes (e.g., color: Red, size: M, quantity: 2)	- Product not added to the cart despite valid inputs	- Cart correctly updates with product details	- Incorrect cart details (e.g., wrong quantity)

5. View cart

Conditions	Valid Partitions	Invalid Partitions	Valid Boundaries	Invalid Boundaries
Cart: Products present	- Cart contains at least one product (e.g., Product A)	- Cart is empty	- Minimum valid cart size = 1 product	- Empty cart with no items to display
	- Cart contains multiple products with correct details (e.g., names, prices, images, quantities)	- Product details are incorrect (e.g., wrong name, mismatched price, missing image)	- Maximum valid cart size (e.g., 50 products)	- Products exceed system capacity (e.g., >50)
Product Details: Displayed correctly	- Product name, price, quantity, and image are displayed for all products	- Missing details like name, price, or image	- Minimal valid detail set (e.g., name + price)	- Missing key details (e.g., no price/image)
Quantity: Adjustable by user	- Positive integer adjustment (e.g., 1 to 5)	- Zero or negative quantity (e.g., 0, -1)	- Minimum valid quantity = 1	- Quantity = 0 or negative

	- Large positive integer adjustment (e.g., 999)	- Non-integer input (e.g., decimal: 1.5, string: One, special chars: @)	- Maximum system-allowed quantity (e.g., 100)	- Exceeds system limit (e.g., >1000)
Image: Displayed or Placeholder	- Product images are displayed correctly	- Missing images replaced by placeholders (e.g., default "Image not available")	- Minimal valid image resolution	- Image file corrupted or missing entirely
	- Images correctly associated with the respective products	- Incorrect or mismatched images (e.g., Product A shows Product B's image)		
Remove Functionality: Working	- Clicking the "Remove" button successfully removes the product	- Clicking the "Remove" button fails with an error (e.g., "Unable to remove item from cart")	- Minimum valid removal (e.g., 1 product)	- Removal failure despite valid conditions
		- System crash or unresponsive button		

6. Place Order

Conditions	Valid Partitions	Invalid Partitions	Valid Boundaries	Invalid Boundaries
Cart: Products available	- Cart contains one or more products	- Cart is empty	- Minimum valid cart size = 1 product	- Empty cart with no items to place order
	- Correct details of products in the cart	- Incorrect or mismatched product details (e.g., wrong quantity, missing price)	- Maximum valid cart size (e.g., 50 products)	- Exceeds cart capacity (e.g., >50 products)

Shipping Details: Provided	- All required fields filled correctly (e.g., address, city, postal code)	- Missing required fields (e.g., no address)	- Minimal valid details (e.g., valid address)	- Missing or invalid details (e.g., blank fields)
	- Valid address format (e.g., 123 Street, City, Postal Code)	- Invalid address (e.g., special characters: @!#123)	- Maximum length for address field (e.g., 255 chars)	- Exceeds address length limit
Payment Options: Selected correctly	- Valid payment method selected (e.g., credit card, PayPal, cash on delivery)	- No payment method selected	- Minimal payment details provided (e.g., valid card number)	- Invalid payment details (e.g., expired card)
	- Payment details entered correctly	- Invalid payment input (e.g., incorrect card number, expired card, insufficient funds)	- Boundary date for card expiration	- Expiration date in the past
Order Confirmation: Valid	- System validates all details and stores the order	- System fails to validate order due to missing or incorrect information	- Successful confirmation message	- Error message: "Unable to process the order"
		- Order not stored in database despite valid input		

7. Manage Profile

Conditions	Valid Partitions	Invalid Partitions	Valid Boundaries	Invalid Boundaries
Username: Editable	- Alphabetic input (e.g., JohnDoe)	- Empty field ("")	- Minimum valid length = 1 character	- Length = 0
	- Alphanumeric input (e.g., John123)	- Exceeds max length (e.g.,	- Maximum valid length = 50 characters	- Special characters only (e.g., @!#)

		>50 characters)		
	- Mixed case input (e.g., JohnDoe, johndoe)	- Numeric only input (e.g., 12345)		
Email: Editable	- Valid email format (e.g., john.doe@gmail.com)	- Invalid email format (e.g., john.doe@)	- Minimum valid length (e.g., a@b.c)	- Missing @ or domain (e.g., johndoe.com)
	- Case-insensitive valid email (e.g., John.Doe@gmail.com, john.doe@gmail.com)	- Exceeds max length (e.g., >100 characters)	- Maximum valid length = 100 characters	- Special characters in invalid positions
Address: Editable	- Valid alphanumeric input (e.g., 123 Street, City)	- Empty field ("")	- Minimum valid input = 5 characters	- Length < 5
	- Includes spaces, commas, and periods (e.g., Apartment #4, Street 5.)	- Special characters only (e.g., !@#\$%^&*)	- Maximum valid input = 255 characters	- Length > 255
Phone Number: Editable	- Valid Pakistani format (e.g., 03001234567)	- Invalid format (e.g., 12345, abcd123)	- Valid length = 11 digits	- Length < 11 or >11 digits
	- Includes leading zero or international code (e.g., +923001234567)	- Non-numeric characters (e.g., 0300abc567)	- Valid country code input (e.g., +92)	- Missing leading zero or country code
Editable Fields: Accessibility	- Editable fields display Save and Cancel buttons upon clicking Edit	- Save and Cancel buttons not displayed	- Buttons responsive to input change	- Buttons unresponsive due to UI bugs
	- Save button updates the field after input	- Field not updated after clicking Save		
Data Validation: System checks input	- Input passes all validation checks (username, email, address, phone number)	- Validation fails for one or more fields	- Minimum valid input for each field	- Combination of invalid inputs

Database Update: Success	- System updates the database with valid inputs	- Database error prevents updating	- Database reflects updated profile	- System fails silently without saving
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8. Add Product (Admin)

Conditions	Valid Partitions	Invalid Partitions	Valid Boundaries	Invalid Boundaries
Product ID: Editable	- Valid alphanumeric ID (e.g., P12345, A001, XYZ999)	- Empty ID field ("")	- Minimum valid length = 1 character	- Length = 0
	- Product ID in the correct format (e.g., a mix of letters and numbers)	- Non-alphanumeric characters (@!#123, *product@)	- Maximum valid length = 10 characters	- Exceeds maximum allowed length (e.g., >10)
Product Name: Editable	- Alphanumeric product name (e.g., Laptop, Smartphone A2021)	- Empty product name field ("")	- Minimum valid length = 1 character	- Length = 0
	- Correct capitalization (e.g., Apple iPhone, Samsung Galaxy)	- Invalid characters (e.g., @!#)	- Maximum valid length = 50 characters	- Exceeds maximum allowed length (e.g., >50)
Price: Editable	- Valid price format (positive decimal, e.g., 19.99, 150.00)	- Negative or zero value (e.g., -10.00, 0.00)	- Minimum valid price = 0.01	- Price = 0 or negative
	- Correct decimal point placement (e.g., 25.5, 100.00)	- Invalid price format (e.g., 25,50, one hundred)	- Maximum price (e.g., 99999.99)	- Exceeds price limit (e.g., >1000000)
Category: Editable	- Valid category name from predefined list (e.g., Electronics, Furniture, Clothing)	- Empty category field ("")	- Minimum valid category length = 1 character	- Length = 0
	- Valid selection from a dropdown menu	- Invalid category name (e.g., special characters @#%,	- Maximum valid length = 50 characters	- Exceeds maximum

	(e.g., Fashion, Books)	non-existent category)		allowed length (e.g., >50)
Description: Editable	- Valid alphanumeric description (e.g., A high-quality laptop with 16GB RAM)	- Empty description field ("")	- Minimum valid length = 5 characters	- Length = 0
	- Full product description with necessary details (e.g., This is a top-rated laptop with a 1TB SSD)	- Invalid description (e.g., excessive special characters or HTML tags <!DOCTYPE html>)	- Maximum valid length = 255 characters	- Exceeds maximum allowed length (e.g., >255)
Quantity: Editable	- Valid positive integer (e.g., 10, 100, 5000)	- Negative quantity (-1, -10)	- Minimum valid quantity = 1	- Quantity = 0 or negative
	- Valid stock number for the product (e.g., 50, 2000)	- Non-integer or decimal value (e.g., 5.5, 100.1)	- Maximum quantity (e.g., 10000)	- Exceeds system stock limit (e.g., >10000)
Product Details Validation	- All fields contain valid data (ID, Name, Price, Category, Description, Quantity)	- One or more invalid fields (e.g., empty fields, invalid characters)	- All required fields are filled	- Missing required fields (e.g., empty product name)
Database Update: Success	- Product is successfully added to the database	- Database update failure (e.g., due to system error)	- Product successfully stored in database	- Error in database update (e.g., transaction failure)
UI Update: Displayed	- The product is correctly displayed in the product listing page	- Product not displayed due to UI or backend errors	- Product appears on product list	- Product fails to display on UI due to errors

9. View Orders (Admin)

Conditions	Valid Partitions	Invalid Partitions	Valid Boundaries	Invalid Boundaries
Order Display	- Orders are visible (e.g., all customer orders displayed)	- No orders displayed (if no orders are placed)	- Minimum valid order = 1 order	- No orders placed
Order Details View	- Admin can view full order details (e.g., items, quantities, customer info)	- Order details missing or not accessible due to errors	- Valid order details shown	- Incomplete or no order details available
Order Status Update	- Admin updates order status (e.g., "shipped", "processing")	- Admin cannot update status (e.g., missing permission, UI error)	- Order status update successful	- No status change after admin action
Order Action Validation	- Admin takes valid actions (update status, cancel order)	- Invalid actions (e.g., canceling an already shipped order)	- Correct action taken	- Invalid action (e.g., cancellation error)

10. Add Reviews for products

Conditions	Valid Partitions	Invalid Partitions	Valid Boundaries	Invalid Boundaries
Product Selection	- Customer has purchased the product and selects it from the order history	- Product not purchased (customer cannot review unpurchased items)	- Product listed in order history	- Product not listed (not purchased)
Rating: 1-5 stars	- Rating between 1 and 5 stars (e.g., 3, 5, 1)	- Rating outside 1-5 stars (e.g., 0, 6)	- Minimum valid rating = 1 star	- Rating = 0 or > 5
Review Title	- Valid title (e.g., Excellent product!, Good value for money)	- Empty title ("")	- Minimum valid length = 1 character	- Title = 0 characters

Review Description	- Valid description (e.g., This product is great, it works well for my needs!)	- Empty description ("") or description with invalid characters (e.g., @#%)	- Minimum valid length = 5 characters	- Description = 0 characters
Submit Review Action	- Customer clicks on Submit Review after completing all fields	- Customer clicks Submit without filling all required fields (e.g., missing title or rating)	- Review successfully submitted	- Review not submitted due to missing data
Review Validation	- System validates all fields and checks for empty inputs or invalid formats	- Invalid or missing fields (e.g., missing rating or description, invalid characters in description)	- All fields correctly filled	- One or more fields missing or incorrect
Review Submission Confirmation	- Successful review submission with confirmation message ("Review submitted successfully")	- Failure to submit review (e.g., due to invalid data or system error)	- Review successfully stored and confirmed	- Submission failure (e.g., database error)
Database Entry	- Review is successfully saved in the database	- Review submission fails (e.g., system/database issues)	- Review saved in the database	- Database error preventing review storage
Review Display on Product Page	- Review is displayed under the product's review section, marked as pending approval	- Review not displayed due to failure in the submission process or database issue	- Review appears under product review section	- Review does not appear under product
Review Status in Admin Panel	- Review appears as "pending approval" in the admin panel	- Review does not appear in the admin panel due to failure in database update or data corruption	- Review status is pending approval	- Review not listed in admin panel

11. Help center request (Customer)

Conditions	Valid Partitions	Invalid Partitions	Valid Boundaries	Invalid Boundaries
Request Type Selection	- Customer selects a valid request type (e.g., "Technical Issue", "Order Query", "Product Inquiry")	- Invalid request type selected (e.g., non-existing options)	- Valid request type selected	- Invalid request type
Subject	- Valid subject (e.g., Issue with order #1234, Product not working as expected)	- Empty subject field ("") or non-descriptive subject (e.g., Help)	- Minimum valid subject length = 1 character	- Empty subject or irrelevant subject
Description	- Valid description (e.g., detailed issue, product inquiry)	- Empty description ("") or non-informative (e.g., Can you help me?)	- Minimum valid description length = 10 characters	- Description = 0 characters
Attachments (Optional)	- Valid attachments (e.g., screenshots, files in supported format like .jpg, .png, .pdf)	- Invalid attachments (e.g., unsupported file formats, empty attachments)	- Valid file format for attachments	- No attachments or unsupported file format
Submit Request Action	- Customer clicks the Submit Request button after filling all fields	- Customer clicks Submit without filling required fields (e.g., missing subject or description)	- Request successfully submitted	- Request not submitted due to incomplete data
Request Validation	- System validates all fields and checks for empty or invalid data	- Missing required fields (subject, description) or invalid attachments	- All fields correctly validated	- Missing required fields or invalid data
Request Confirmation	- System sends a confirmation email and displays a success	- Failure to send confirmation or display success	- Confirmation received	- No confirmation due to

	message ("Your request has been submitted")	message due to system error		submission failure
Database Entry	- Request is saved successfully in the database	- Request fails to save in the database due to system/database error	- Request successfully saved	- Database error preventing request storage
Request Display in Admin Dashboard	- Request is visible in the Admin Dashboard with all details	- Request not displayed in Admin Dashboard due to data issue or failure in the system update	- Request appears with all details	- Request does not appear in the admin dashboard

12. Send complaint (Staff)

Conditions	Valid Partitions	Invalid Partitions	Valid Boundaries	Invalid Boundaries
Complaint Type Selection	- Staff selects a valid complaint type (e.g., "System Error", "Miscommunication", "Lack of Resources")	- Invalid complaint type selected (e.g., non-existing options, blank selection)	- Valid complaint type selected	- No selection or invalid complaint type
Subject	- Valid subject (e.g., Issue with system, Order communication error)	- Empty subject field ("") or non-specific subject (e.g., Help)	- Minimum valid subject length = 1 character	- Empty or irrelevant subject
Description	- Valid description (e.g., detailed complaint explanation)	- Empty description ("") or vague, uninformative description (e.g., There is an issue)	- Minimum valid description length = 10 characters	- Description = 0 characters
Submit Complaint Action	- Staff clicks Submit Complaint after filling all fields	- Staff clicks Submit without filling required fields (e.g., missing subject or description)	- Complaint successfully submitted	- Complaint not submitted due to missing data

Complaint Validation	- System validates all fields and checks for empty or invalid data	- Missing required fields (subject, description) or invalid data (e.g., unsupported characters)	- All fields correctly validated	- Missing or invalid fields
Complaint Submission Confirmation	- System displays a confirmation message ("Complaint submitted successfully")	- Failure to display confirmation message due to system or data errors	- Confirmation received	- No confirmation message
Database Entry	- Complaint is successfully stored in the database	- Complaint fails to be saved in the database due to system/database error	- Complaint stored with unique ID	- Database error preventing complaint storage
Admin Dashboard Visibility	- Complaint appears in the admin's Complaint Management section	- Complaint does not appear in the admin section due to failure in database update or system issue	- Complaint visible with all details	- Complaint not visible in admin dashboard

BVA

1. User Registration

Test Case ID	Objective	Pre-conditions	Steps to Execute	Expected Results	Actual Results	Status (Pass/Fail)
TC-001	Validate username with minimum length	System allows username input	1. Enter a username with a single character (e.g., J). 2. Click the submit/register button.	Username is accepted successfully .	Username accepted successfully .	Pass
TC-002	Validate username with	System allows username input	1. Enter a username with exactly 50 characters. 2. Click	Username is accepted successfully .	Username accepted successfully .	Pass

	maximum length		the submit/register button.			
TC-003	Validate username exceeding max length	System allows username input	1. Enter a username with 51 characters. 2. Click the submit/register button.	System rejects the username and displays an error message: "Username too long."	Error displayed: "Username too long."	Pass
TC-004	Validate empty username	System allows username input	1. Leave the username field empty. 2. Click the submit/register button.	System rejects the username and displays an error message: "Username is required."	Error displayed: "Username is required."	Pass
TC-005	Validate minimally valid email	System allows email input	1. Enter a minimally valid email (e.g., a@gmail.com). 2. Click submit/register.	Email is accepted successfully .	Email accepted successfully .	Pass
TC-006	Validate missing '@' in email	System allows email input	1. Enter an email without the @ symbol (e.g., usergmail.com). 2. Click submit/register.	System rejects the email and displays an error: "Invalid email format."	Error displayed: "Invalid email format."	Pass
TC-007	Validate email with missing domain	System allows email input	1. Enter an email with a missing domain (e.g., user@). 2. Click submit/register.	System rejects the email and displays an error: "Invalid email format."	Error displayed: "Invalid email format."	Pass

TC-008	Validate email with extra characters	System allows email input	1. Enter an email with invalid characters (e.g., user@@gmail.com) . 2. Click submit/register.	System rejects the email and displays an error: "Invalid email format."	Error displayed: "Invalid email format."	Pass
TC-009	Validate password with minimum length	System allows password input	1. Enter a password with exactly 8 characters (e.g., A1@bcdEF). 2. Click submit/register.	Password is accepted successfully .	Password accepted successfully .	Pass
TC-010	Validate password below minimum length	System allows password input	1. Enter a password with 7 characters (e.g., Abc@12). 2. Click submit/register.	System rejects the password and displays an error: "Password too short."	Error displayed: "Password too short."	Pass
TC-011	Validate password exceeding max length	System allows password input	1. Enter a password with 51 characters. 2. Click submit/register.	System rejects the password and displays an error: "Password too long."	Error displayed: "Password too long."	Pass
TC-012	Validate password missing criteria	System allows password input	1. Enter a password missing uppercase letters (e.g., abc@1234). 2. Click submit/register.	System rejects the password and displays an error: "Password must include an uppercase letter."	Error displayed: "Password must include an uppercase letter."	Pass
TC-013	Validate phone number	System allows phone	1. Enter a valid phone number (e.g.,	Phone number is accepted	Phone number accepted	Pass

	with valid length	number input	03012345678). 2. Click submit/register.	successfully .	successfully .	
TC-014	Validate phone number below valid length	System allows phone number input	1. Enter a phone number with 10 digits (e.g., 0312345678). 2. Click submit/register.	System rejects the phone number and displays an error: "Phone number must be 11 digits."	Error displayed: "Phone number must be 11 digits."	Pass
TC-015	Validate phone number with extra chars	System allows phone number input	1. Enter a phone number with special characters (e.g., 0301-2345678). 2. Click submit/register.	System rejects the phone number and displays an error: "Phone number must be numeric only."	Error displayed: "Phone number must be numeric only."	Pass
TC-016	Validate address with valid input	System allows address input	1. Enter a valid address (e.g., 123 Main St). 2. Click submit/register.	Address is accepted successfully .	Address accepted successfully .	Pass
TC-017	Validate empty address field	System allows address input	1. Leave the address field empty. 2. Click submit/register.	Address is accepted successfully (optional field).	Address accepted successfully .	Pass
TC-018	Validate minimal valid address	System allows address input	1. Enter a single character as address (e.g., A). 2. Click submit/register.	Address is accepted successfully .	Address accepted successfully .	Pass

TC-019	Validate long valid address	System allows address input	1. Enter a detailed address (e.g., House #10, Lane 4, Block C, Karachi, Pakistan). 2. Click submit/register.	Address is accepted successfully	Address accepted successfully	Pass
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2. User Login

Test Case ID	Objective	Pre-conditions	Steps to Execute	Expected Results	Actual Results	Status (Pass/Fail)
TC-001	Validate registered email with valid format	System has a registered email	1. Enter a registered email (e.g., user@gmail.com). 2. Submit the login form.	System accepts the email successfully.	System accepts the email successfully.	Pass
TC-002	Validate unregistered email	System does not have the email registered	1. Enter an unregistered email (e.g., newuser@gmail.com). 2. Submit the login form.	System rejects the email and displays an error: "Email not registered."	System rejects the email and displays an error: "Email not registered."	Pass
TC-003	Validate email without '@' symbol	System allows email input	1. Enter an email without the '@' symbol (e.g., usergmail.com). 2. Submit the login form.	System rejects the email and displays an error: "Invalid email format."	System rejects the email and displays an error: "Invalid email format."	Pass

TC-004	Validate email with maximum valid length	System allows email input	1. Enter an email with 50 characters (e.g., verylongemail@exampleexample.com). 2. Submit.	Email is accepted successfully.	Email is accepted successfully.	Pass
TC-005	Validate email exceeding max length	System allows email input	1. Enter an email exceeding 50 characters. 2. Submit the login form.	System rejects the email and displays an error: "Email exceeds allowed length."	System rejects the email and displays an error: "Email exceeds allowed length."	Pass
TC-006	Validate correct password for registered email	System has a registered account	1. Enter a registered email. 2. Enter the correct password associated with the email. Submit.	System logs the user in successfully.	System logs the user in successfully.	Pass
TC-007	Validate incorrect password	System has a registered account	1. Enter a registered email. 2. Enter an incorrect password. Submit.	System rejects the login and displays an error: "Incorrect password."	System rejects the login and displays an error: "Incorrect password."	Pass
TC-008	Validate password with	System allows password input	1. Enter a registered email. 2. Enter a password with exactly 8 characters, meeting criteria. Submit.	Password is accepted	Password is accepted	Pass

	minimal valid length			successfully.	successfully.	
TC-009	Validate password with invalid length (<8)	System allows password input	1. Enter a registered email. 2. Enter a password with fewer than 8 characters (e.g., Ab@12). Submit.	System rejects the password and displays an error: "Password too short."	System rejects the password and displays an error: "Password too short."	Pass
TC-010	Validate long valid password	System allows password input	1. Enter a registered email. 2. Enter a password with 50 characters. Submit.	Password is accepted successfully.	Password is accepted successfully.	Pass
TC-011	Validate empty password	System allows password input	1. Enter a registered email. 2. Leave the password field empty. Submit.	System rejects the login and displays an error: "Password is required."	System rejects the login and displays an error: "Password is required."	Pass

3. Browse product by search

Test Case ID	Objective	Pre-conditions	Steps to Execute	Expected Results	Actual Results	Status (Pass/Fail)
TC-001	Validate exact product name	System has the product in the database	1. Enter the exact product name (e.g., Laptop). 2. Click the Search button.	The system displays the product in the search results.	Product displayed successfully in search results.	Pass

TC-002	Validate search with partial product name	System has products matching the term	1. Enter a partial product name (e.g., Lap). 2. Click the Search button.	The system displays products that match the partial name.	Products matching "Lap" displayed successfully.	Pass
TC-003	Validate search with special characters	System allows input in search field	1. Enter special characters only (e.g., @\$%^). 2. Click the Search button.	The system displays an error: "No results found."	Error message "No results found" displayed as expected.	Pass
TC-004	Validate search with minimum valid input	System allows input in search field	1. Enter a single character (e.g., A). 2. Click the Search button.	The system performs the search and returns relevant results.	Relevant results displayed successfully.	Pass
TC-005	Validate search exceeding max length	System allows input in search field	1. Enter a product name exceeding 50 characters. 2. Click the Search button.	The system displays an error: "Search input exceeds the allowed length."	Error message "Search input exceeds the allowed length" displayed.	Pass
TC-007	Validate empty search input	System allows input in search field	1. Leave the search field empty. 2. Click the Search button.	The system displays an error: "Search field cannot be empty."	Error message "Search field cannot be empty" displayed.	Pass
TC-008	Validate results for an existing product	System has products matching the search term	1. Search for a product name (e.g., Laptop). 2. View the results.	Products matching the term are displayed.	Products matching "Laptop" displayed successfully.	Pass
TC-009	Validate no results for non-	System has no	1. Search for a non-existent product name (e.g.,	The system displays an error: "No	Error message "No results found"	Pass

	existent product	matching products	UnicornLaptop). 2. View the results.	results found."	displayed as expected.	
TC-012	Validate correct display of all details	System has detailed information for products	1. Click on a product from the search results. 2. View the product details page.	The page displays name, price, description, and dropdown options for attributes.	Product details displayed successfully with all attributes.	Pass
TC-014	Validate dropdown options for attributes	Product has selectable attributes	1. Click on a product. 2. Verify dropdown options (e.g., size, color) are displayed and selectable.	Dropdowns for attributes are functional and contain correct values.	Dropdown options displayed and functional.	Pass
TC-015	Validate missing product details	Product has incomplete information	1. Click on a product with missing details. 2. View the product details page.	The page displays a placeholder or message: "Details not available."	Placeholder message "Details not available" displayed.	Pass
TC-016	Validate successful navigation to product details	Product exists in the system	1. Click on a product from the search results. 2. Observe the redirection to the product details page.	The product details page loads within < 2 seconds.	Product details page loaded successfully within 2 seconds.	Pass
TC-017	Validate slow or failed navigation	System experiences delays	1. Click on a product from the search results. 2. Observe the system response.	The system displays an error: "Unable to load product details."	Error message "Unable to load product details" displayed.	Pass

4. Add Product to cart

Test Case ID	Objective	Pre-conditions	Steps to Execute	Expected Results	Actual Results	Status (Pass/Fail)
TC-001	Validate all required dropdowns selected	Dropdowns are functional	1. Select valid options for all required dropdowns (e.g., color: Red, size: M).	The system allows proceeding to the next step or adding to cart.	Dropdown options selected; product successfully added to the cart.	Pass
TC-002	Validate missing required dropdown values	Dropdowns are functional	1. Leave one or more required dropdowns unselected. 2. Attempt to proceed or add to cart.	The system displays an error: "Please select all required options."	Error message displayed: "Please select all required options."	Pass
TC-003	Validate invalid values in dropdowns	Dropdowns accept user selection	1. Attempt to select an invalid or non-existent option (e.g., size: Pink).	The system prevents selection or displays an error message.	Invalid option not selectable; error message displayed.	Pass
TC-004	Validate positive integer quantity	System allows quantity input	1. Enter a valid positive integer for quantity (e.g., 1, 5). 2. Click Add to Cart.	The product is added to the cart with the specified quantity.	Product successfully added with the specified quantity.	Pass
TC-005	Validate minimum valid quantity	System allows quantity input	1. Enter the minimum valid quantity (e.g., 1).	The product is added to the cart	Product successfully added with quantity 1.	Pass

			2. Click Add to Cart.	with quantity 1.		
TC-006	Validate invalid quantity (zero or negative)	System allows quantity input	1. Enter an invalid quantity (e.g., 0, -1). 2. Attempt to add to cart.	The system displays an error: "Quantity must be a positive integer."	Error message displayed: "Quantity must be a positive integer."	Pass
TC-007	Validate button is clickable with valid inputs	All inputs are valid	1. Select valid dropdown options. 2. Enter a valid quantity. 3. Click Add to Cart.	The product is added to the cart, and a success message is displayed.	Product successfully added; success message displayed.	Pass
TC-008	Validate button is unresponsive with missing inputs	Dropdowns or quantity are invalid	1. Leave one or more required fields incomplete. 2. Attempt to click Add to Cart.	The button remains inactive or an error message is displayed.	Error message displayed: "Please complete all required fields."	Pass
TC-009	Validate success message on valid addition	All inputs are valid	1. Complete all required fields. 2. Click Add to Cart.	A success message is displayed: "Product added to cart."	Success message displayed: "Product added to cart."	Pass
TC-010	Validate error message on invalid inputs	One or more inputs are invalid	1. Leave a required field incomplete. 2. Click Add to Cart.	An error message is displayed: "Please select all required options."	Error message displayed: "Please select all required options."	Pass

TC-011	Validate cart reflects correct product details	Product added successfully	1. Add a product with valid attributes to the cart. 2. Open the cart and review the product details.	The cart displays the correct product name, quantity, and selected attributes.	Cart displays correct product name, quantity, and selected attributes.	Pass
TC-012	Validate cart does not reflect incorrect details	Product added unsuccessfully	1. Add a product with missing or incorrect inputs. 2. Open the cart and review the product details.	The product is not added to the cart, or incorrect details are flagged.	Product not added to cart; system prevented incorrect input addition.	Pass

5. View cart

Test Case ID	Objective	Pre-conditions	Steps to Execute	Expected Results	Actual Results	Status (Pass/Fail)
TC-001	Validate cart with at least one product	Cart contains products	1. Add at least one product to the cart. 2. View the cart.	The cart displays the product(s) with correct details (name, price, image, quantity).	The cart displayed all added products with correct details.	Pass
TC-002	Validate empty cart behavior	Cart is empty	1. Remove all products from the cart. 2. View the cart.	The cart displays a message: "Your cart is empty."	The cart displayed the message: "Your cart is empty."	Pass
TC-003	Validate correct product details in cart	Products are added to the cart	1. Add multiple products to the cart. 2. Verify	All product details are displayed correctly.	All product details were displayed accurately.	Pass

			product details (name, price, quantity, image).			
TC-004	Validate missing product details	Products are added to the cart	1. Add products with missing or incorrect details. 2. View the cart.	Missing or incorrect details are flagged or handled appropriately.	Missing details were flagged, and a warning was shown.	Pass
TC-005	Validate quantity adjustment with valid values	Cart contains at least one product	1. Change the quantity of a product to a valid positive integer. 2. Save or confirm the change.	The quantity updates correctly in the cart, and the total price is recalculated.	Quantity updated correctly, and total price was recalculated accurately.	Pass
TC-006	Validate invalid quantity adjustment	Cart contains at least one product	1. Change the quantity to an invalid value (e.g., 0, -1, non-integer). 2. Save or confirm the change.	The system displays an error: "Invalid quantity entered."	Error message: "Invalid quantity entered" was displayed as expected.	Pass
TC-007	Validate maximum cart size	Cart is populated with products	1. Add the maximum allowable number of products to the cart. 2. Attempt to add more products.	The system prevents adding more than the maximum limit (e.g., 50 products).	System correctly prevented adding more than the allowed limit.	Pass
TC-008	Validate product image display	Cart contains products with images	1. Add products with images to the cart. 2. Verify the displayed	Images are correctly displayed for respective products, or placeholders are shown.	Correct images were displayed for all products.	Pass

			images for correctness.			
TC-009	Validate mismatched product images	Cart contains products with images	1. Add products with incorrect image associations. 2. View the cart.	Mismatched images are flagged or handled appropriately.	Mismatched images were flagged, and error messages were shown.	Pass
TC-010	Validate product removal functionality	Cart contains at least one product	1. Click the "Remove" button for a product. 2. Verify if the product is removed from the cart.	The product is removed successfully, and the cart updates correctly.	Product was successfully removed, and cart updated as expected.	Pass
TC-011	Validate failure to remove product from cart	Cart contains at least one product	1. Click the "Remove" button for a product under error conditions. 2. Observe the behavior.	The system displays an error: "Unable to remove item from cart."	Error message: "Unable to remove item from cart" was displayed as expected.	Pass

6. Place Order

Test Case ID	Objective	Pre-conditions	Steps to Execute	Expected Results	Actual Results	Status (Pass/Fail)
TC-001	Validate cart with at least one product	Cart contains one or more products	1. Add at least one product to the cart. 2. View the cart.	The cart displays the product(s) with correct details (name, price, quantity).	Cart displayed product(s) with correct details (name, price, quantity).	Pass
TC-002	Validate behavior for	Cart is empty	1. Remove all products from the cart.	System prevents checkout and displays a	System displayed "Your cart is empty" and	Pass

	an empty cart		2. Attempt to proceed to checkout.	message: "Your cart is empty."	prevented checkout.	
TC-003	Validate correct product details in cart	Cart contains one or more products	1. Add multiple products to the cart. 2. Verify product details (name, price, quantity).	All product details are displayed correctly.	Product details (name, price, quantity) were displayed correctly.	Pass
TC-004	Validate mismatched or missing product details	Cart contains products	1. Add products with incomplete or incorrect details. 2. View the cart.	Missing or incorrect details are flagged or handled appropriately.	System flagged missing details with a "Data missing" warning.	Pass
TC-005	Validate cart size with maximum products	Cart is populated with products	1. Add the maximum allowable number of products to the cart. 2. Attempt to add more products.	The system prevents adding more than the maximum limit (e.g., 50 products).	System prevented adding more than 50 products.	Pass
TC-006	Validate shipping details with correct input	Cart contains one or more products	1. Proceed to checkout. 2. Fill in all required shipping details. 3. Submit the form.	System accepts the input and navigates to payment options.	System accepted details and navigated to payment options.	Pass

TC-007	Validate missing shipping details	Cart contains one or more products	1. Proceed to checkout. 2. Leave required fields blank. 3. Submit the form.	System prevents submission and displays an error: "Please fill in all required fields."	Error message displayed: "Please fill in all required fields."	Pass
TC-008	Validate invalid shipping address	Cart contains one or more products	1. Proceed to checkout. 2. Enter an invalid address format (e.g., @!#123). 3. Submit the form.	System prevents submission and displays an error: "Invalid address format."	System displayed "Invalid address format" error message.	Pass
TC-009	Validate maximum shipping address length	Cart contains one or more products	1. Proceed to checkout. 2. Enter a long valid address (e.g., 255 chars). 3. Submit the form.	System accepts the input and navigates to payment options.	System accepted input and navigated to payment options.	Pass
TC-010	Validate valid payment method selection	Cart contains one or more products	1. Fill in valid shipping details. 2. Select a valid payment method. 3. Submit the payment.	System processes the payment and displays a success message.	Payment processed successfully, success message displayed.	Pass

TC-011	Validate missing payment method	Cart contains one or more products	1. Fill in valid shipping details. 2. Do not select a payment method. 3. Submit the form.	System prevents submission and displays an error: "Please select a payment method."	Error message displayed: "Please select a payment method."	Pass
TC-012	Validate expired payment method	Cart contains one or more products	1. Fill in valid shipping details. 2. Enter payment details with an expired card. 3. Submit.	System prevents submission and displays an error: "Payment method expired."	System displayed "Payment method expired" error message.	Pass
TC-013	Validate order confirmation with valid details	Cart contains one or more products	1. Provide valid shipping and payment details. 2. Confirm the order.	System validates details, stores the order, and displays a success message.	Order was stored, and success message displayed.	Pass
TC-014	Validate order failure due to invalid details	Cart contains one or more products	1. Provide invalid shipping or payment details. 2. Confirm the order.	System displays an error: "Unable to process the order" and prevents confirmation.	Error displayed: "Unable to process the order."	Pass

7. Manage Profile

Test Case ID	Objective	Pre-conditions	Steps to Execute	Expected Results	Actual Results	Status (Pass/Fail)
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TC-001	Validate username field with valid alphabetic input	Username field is editable	1. Enter alphabetic username (e.g., JohnDoe). 2. Click Save.	System accepts the input and updates the username in the database.	System accepts the input and updates the username in the database.	Pass
TC-002	Validate username field with empty input	Username field is editable	1. Leave the username field empty. 2. Click Save.	System prevents save and displays an error: "Username cannot be empty."	System prevents save and displays an error: "Username cannot be empty."	Pass
TC-003	Validate username field with valid alphanumeric input	Username field is editable	1. Enter alphanumeric username (e.g., John123). 2. Click Save.	System accepts the input and updates the username in the database.	System accepts the input and updates the username in the database.	Pass
TC-004	Validate username field exceeding max length	Username field is editable	1. Enter a username with more than 50 characters. 2. Click Save.	System prevents save and displays an error: "Username exceeds maximum length."	System prevents save and displays an error: "Username exceeds maximum length."	Pass
TC-005	Validate email field with valid format	Email field is editable	1. Enter a valid email (e.g., john.doe@gmail.com). 2. Click Save.	System accepts the input and updates the email	System accepts the input and updates the email	Pass

				in the database.	in the database.	
TC-006	Validate email field with invalid format	Email field is editable	1. Enter an invalid email format (e.g., john.doe@). 2. Click Save.	System prevents save and displays an error: "Invalid email format."	System prevents save and displays an error: "Invalid email format."	Pass
TC-007	Validate email field with max length	Email field is editable	1. Enter an email with more than 100 characters. 2. Click Save.	System prevents save and displays an error: "Email exceeds maximum length."	System prevents save and displays an error: "Email exceeds maximum length."	Pass
TC-008	Validate address field with valid input	Address field is editable	1. Enter a valid address (e.g., 123 Street, City). 2. Click Save.	System accepts the input and updates the address in the database.	System accepts the input and updates the address in the database.	Pass
TC-009	Validate address field with empty input	Address field is editable	1. Leave the address field empty. 2. Click Save.	System prevents save and displays an error: "Address cannot be empty."	System prevents save and displays an error: "Address cannot be empty."	Pass
TC-010	Validate address field with invalid characters	Address field is editable	1. Enter an address with special characters only (e.g., !@#\$%^&*). 2. Click Save.	System prevents save and displays an error:	System prevents save and displays an error:	Pass

				"Invalid address format."	"Invalid address format."	
TC-011	Validate phone number field with valid Pakistani format	Phone number field is editable	1. Enter a valid Pakistani phone number (e.g., 03001234567). 2. Click Save.	System accepts the input and updates the phone number in the database.	System accepts the input and updates the phone number in the database.	Pass
TC-012	Validate phone number field with invalid format	Phone number field is editable	1. Enter an invalid phone number format (e.g., abcd123). 2. Click Save.	System prevents save and displays an error: "Invalid phone number format."	System prevents save and displays an error: "Invalid phone number format."	Pass
TC-013	Validate phone number field with incorrect length	Phone number field is editable	1. Enter a phone number with less than or more than 11 digits. 2. Click Save.	System prevents save and displays an error: "Phone number must be 11 digits."	System prevents save and displays an error: "Phone number must be 11 digits."	Pass
TC-014	Validate editable fields displaying Save/Cancel buttons	Editable fields are shown	1. Click Edit on any editable field (username, email, address, phone). 2. Verify Save/Cancel buttons.	Save and Cancel buttons appear and are functional .	Save and Cancel buttons appear and are functional .	Pass
TC-015	Validate Save button after valid input	Editable fields are shown	1. Edit any field with valid input. 2. Click Save.	System saves the changes and	System saves the changes and	Pass

				updates the database.	updates the database.	
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8. Add Product (Admin)

Test Case ID	Objective	Pre-conditions	Steps to Execute	Expected Results	Actual Results	Status (Pass/Fail)
TC-001	Validate Product ID with valid alphanumeric input	Product ID field is editable	1. Enter valid alphanumeric Product ID (e.g., P12345). 2. Click Save.	System accepts the valid Product ID and updates the database.	System accepted the Product ID and updated the database.	Pass
TC-002	Validate Product ID with empty input	Product ID field is editable	1. Leave the Product ID field empty. 2. Click Save.	System prevents save and displays an error: "Product ID cannot be empty."	System displayed error: "Product ID cannot be empty."	Pass
TC-003	Validate Product ID exceeding max length	Product ID field is editable	1. Enter a Product ID with more than 10 characters. 2. Click Save.	System prevents save and displays an error: "Product ID exceeds maximum length."	System displayed error: "Product ID exceeds maximum length."	Pass
TC-004	Validate Product Name with valid alphanumeric input	Product Name field is editable	1. Enter valid product name (e.g., Laptop). 2. Click Save.	System accepts the valid product name and updates the database.	System accepted the product name and updated the database.	Pass

TC-005	Validate Product Name with invalid characters	Product Name field is editable	1. Enter product name with invalid characters (e.g., @!#). 2. Click Save.	System prevents save and displays an error: "Invalid product name."	System displayed error: "Invalid product name."	Pass
TC-006	Validate Product Name exceeding max length	Product Name field is editable	1. Enter a Product Name exceeding 50 characters. 2. Click Save.	System prevents save and displays an error: "Product Name exceeds maximum length."	System displayed error: "Product Name exceeds maximum length."	Pass
TC-007	Validate Price with valid decimal input	Price field is editable	1. Enter valid price (e.g., 19.99). 2. Click Save.	System accepts the valid price and updates the database.	System accepted the price and updated the database.	Pass
TC-008	Validate Price with negative or zero value	Price field is editable	1. Enter negative price (e.g., -10.00). 2. Click Save.	System prevents save and displays an error: "Price cannot be negative or zero."	System displayed error: "Price cannot be negative or zero."	Pass
TC-009	Validate Price with invalid format	Price field is editable	1. Enter price in an invalid format (e.g., 25,50). 2. Click Save.	System prevents save and displays an error: "Invalid price format."	System displayed error: "Invalid price format."	Pass

TC-010	Validate Price exceeding maximum allowed value	Price field is editable	1. Enter price exceeding allowed value (e.g., 1000000). 2. Click Save.	System prevents save and displays an error: "Price exceeds maximum allowed value."	System displayed error: "Price exceeds maximum allowed value."	Pass
TC-011	Validate Category with valid input from list	Category field is editable	1. Select valid category from dropdown (e.g., Electronics). 2. Click Save.	System accepts the valid category and updates the database.	System accepted the category and updated the database.	Pass
TC-012	Validate Category with invalid characters	Category field is editable	1. Enter an invalid category (e.g., @#%). 2. Click Save.	System prevents save and displays an error: "Invalid category name."	System displayed error: "Invalid category name."	Pass
TC-013	Validate Category exceeding max length	Category field is editable	1. Enter category with more than 50 characters. 2. Click Save.	System prevents save and displays an error: "Category name exceeds maximum length."	System displayed error: "Category name exceeds maximum length."	Pass
TC-014	Validate Description with valid alphanumeric input	Description field is editable	1. Enter a valid description (e.g., High-quality laptop with 16GB RAM). 2. Click Save.	System accepts the valid description and updates the database.	System accepted the description and updated the database.	Pass

TC-015	Validate Description with empty input	Description field is editable	1. Leave the description field empty. 2. Click Save.	System prevents save and displays an error: "Description cannot be empty."	System displayed error: "Description cannot be empty."	Pass
TC-016	Validate Description with excessive special characters	Description field is editable	1. Enter an invalid description (e.g., excessive HTML tags). 2. Click Save.	System prevents save and displays an error: "Invalid description format."	System displayed error: "Invalid description format."	Pass
TC-017	Validate Description exceeding max length	Description field is editable	1. Enter a description exceeding 255 characters. 2. Click Save.	System prevents save and displays an error: "Description exceeds maximum length."	System displayed error: "Description exceeds maximum length."	Pass
TC-018	Validate Quantity with valid positive integer	Quantity field is editable	1. Enter valid quantity (e.g., 100). 2. Click Save.	System accepts the valid quantity and updates the database.	System accepted the quantity and updated the database.	Pass
TC-019	Validate Quantity with negative value	Quantity field is editable	1. Enter negative quantity (e.g., -10). 2. Click Save.	System prevents save and displays an error: "Quantity cannot be negative."	System displayed error: "Quantity cannot be negative."	Pass

TC-020	Validate Quantity with decimal value	Quantity field is editable	1. Enter a decimal quantity (e.g., 10.5). 2. Click Save.	System prevents save and displays an error: "Quantity must be an integer."	System displayed error: "Quantity must be an integer."	Pass
TC-021	Validate Quantity exceeding system limit	Quantity field is editable	1. Enter quantity exceeding system limit (e.g., 10001). 2. Click Save.	System prevents save and displays an error: "Quantity exceeds system limit."	System displayed error: "Quantity exceeds system limit."	Pass

9. View Orders (Admin)

Test Case ID	Objective	Pre-conditions	Steps to Execute	Expected Results	Actual Results	Status (Pass/Fail)
TC-001	Validate Order Display with valid orders	Orders are placed in the system	1. Ensure there are valid orders placed. 2. View the order list.	All customer orders are displayed.	All customer orders are displayed correctly.	Pass
TC-002	Validate Order Display when no orders are placed	No orders are placed in the system	1. Ensure no orders are placed. 2. View the order list.	No orders are displayed (empty order list).	No orders are displayed (empty order list).	Pass
TC-003	Validate Order Details View for full order information	Orders are placed and accessible	1. Click on an order. 2. Verify the display of full order details (items, quantities,	All order details are visible, including items, quantities,	All order details are visible, including items and quantities.	Pass

			customer info).	and customer information.		
TC-004	Validate Order Details View with missing information	Orders placed but some details are missing	1. Click on an order with incomplete details. 2. Verify missing order details.	System displays an error or missing details message (e.g., incomplete order).	System displays missing details message.	Pass
TC-005	Validate Order Status Update by Admin	Admin has proper permissions to update status	1. Select an order. 2. Admin updates the order status (e.g., "shipped" or "processing").	The order status is successfully updated.	Order status updated to "shipped".	Pass
TC-006	Validate inability to update order status without permission	Admin lacks proper permissions	1. Try updating the order status without sufficient permissions. 2. Attempt the status update.	System prevents status update and shows an error: "Permission denied."	Status update prevented with error: "Permission denied."	Pass
TC-007	Validate Order Status Update when action fails	Order update process is initiated	1. Try updating the order status (e.g., "shipped"). 2. Verify if the status change is applied correctly.	Status update is successful and reflected in the system.	Status update failed due to system error.	Fail
TC-008	Validate inability to update order	UI error present during status update	1. Attempt to update the status. 2. Observe any UI errors	UI error occurs and status is not updated.	UI error occurred and status not updated.	Fail

	status due to UI error		during status update.			
TC-009	Validate Admin actions (e.g., update status, cancel order)	Admin has the necessary permissions	1. Admin updates the order status (e.g., "shipped"). 2. Admin cancels an order (if applicable).	The correct action is applied (status update or order cancellation).	Admin successfully updated status and cancelled order.	Pass
TC-010	Validate invalid action (e.g., canceling a shipped order)	Order status is already "shipped"	1. Admin attempts to cancel an already shipped order. 2. Observe the system behavior.	System prevents invalid action and shows error: "Cannot cancel a shipped order."	System prevented cancellation with error: "Cannot cancel."	Pass
TC-011	Validate invalid action (e.g., incorrect status update)	Admin attempts invalid actions	1. Admin attempts an invalid action, such as updating to a non-existent status. 2. Observe the system.	System shows an error message: "Invalid status update."	System displayed error: "Invalid status update."	Pass

10. Add Reviews for products

Test Case ID	Objective	Pre-conditions	Steps to Execute	Expected Results	Actual Results	Status (Pass/Fail)
TC-001	Validate Product Selection from Order History	Customer has purchased the product	1. Customer logs in. 2. Selects a product from the order history.	Product is listed in the order history.	Product listed successfully in order history.	Pass

			3. Ensure product is listed.			
TC-002	Validate Invalid Product Selection (Not Purchased)	Customer has not purchased the product	1. Customer tries to select an unpurchased product from order history. 2. Observe the system behavior.	Product is not listed in the order history (error message displayed).	Error message displayed: "Product not purchased".	Pass
TC-003	Validate Rating Submission within Valid Range	Customer is rating a product	1. Customer selects a rating between 1 and 5 stars. 2. Submit the rating.	Rating is submitted successfully (between 1 and 5 stars).	Rating submitted successfully (4 stars).	Pass
TC-004	Validate Rating Submission Outside Valid Range	Customer is rating a product	1. Customer selects a rating outside 1-5 stars (e.g., 0 or 6). 2. Attempt to submit the rating.	Rating submission fails (error message displayed for invalid rating).	Error message: "Invalid rating, must be between 1-5 stars".	Pass
TC-005	Validate Review Title Submission	Customer enters a valid title	1. Customer enters a valid title (e.g., "Excellent product!"). 2. Submit the review.	Review submitted successfully with a valid title.	Review submitted with valid title.	Pass
TC-006	Validate Empty or Invalid Review Title	Customer leaves the title empty or invalid	1. Customer leaves the title empty or enters	Review submission fails (error message	Error message: "Title cannot be	Pass

			invalid characters. 2. Attempt to submit the review.	displayed for invalid or empty title).	empty or invalid."	
TC-007	Validate Review Description Submission	Customer enters a valid description	1. Customer enters a valid description (e.g., "This product works great!"). 2. Submit the review.	Review submitted successfully with a valid description.	Review submitted with valid description.	Pass
TC-008	Validate Empty or Invalid Review Description	Customer leaves description empty or invalid	1. Customer leaves the description empty or enters invalid characters. 2. Attempt to submit the review.	Review submission fails (error message displayed for invalid or empty description).	Error message: "Description cannot be empty or invalid."	Pass
TC-009	Validate Submit Review Action	All required fields are filled	1. Customer clicks on "Submit Review" after completing all fields. 2. Review is submitted.	Review is successfully submitted.	Review successfully submitted.	Pass
TC-010	Validate Submit Review Action with Missing Fields	Missing required fields (e.g., title or rating)	1. Customer attempts to submit the review with missing fields. 2. Observe	Review submission fails (error message displayed for missing fields).	Error message: "Title or rating is missing."	Pass

			the system behavior.			
TC-011	Validate Review Validation	System validates all fields	1. System validates the review fields (title, rating, description) 2. Observe any validation errors.	Review is validated successfully (all fields correctly filled).	All fields validated successfully.	Pass
TC-012	Validate Review Submission Confirmation	Review is submitted	1. Review is successfully submitted. 2. Verify the confirmation message is shown.	Review submission confirmation message ("Review submitted successfully").	Confirmation message displayed.	Pass
TC-013	Validate Review Submission Failure	Invalid data or system error	1. Customer submits invalid data or encounters a system error. 2. Observe the failure behavior.	Submission fails (error message displayed for invalid data or system error).	Error message displayed for invalid data.	Pass
TC-014	Validate Database Entry for Review	Review is submitted	1. Submit a valid review. 2. Verify the review is saved in the database.	Review is saved in the database.	Review saved in the database.	Pass
TC-015	Validate Review Display on	Review is successfully submitted	1. View the product page where the review	Review appears under the product's	Review displayed under	Pass

	Product Page		was submitted. 2. Ensure the review is displayed under the product.	review section, marked as "pending approval".	product page.	
TC-016	Validate Review Display Failure on Product Page	Review submission fails or there is a database issue	1. View the product page. 2. Ensure the review is not displayed due to submission failure.	Review is not displayed due to failure in the submission process or database issue.	Review not displayed due to submission failure.	Pass
TC-017	Validate Review Status in Admin Panel	Review is successfully submitted	1. Admin accesses the admin panel. 2. Check the review status for the submitted review.	Review status is displayed as "pending approval" in the admin panel.	Review status: "pending approval".	Pass
TC-018	Validate Review Not Displayed in Admin Panel due to Error	System/database error prevents review entry	1. Check the admin panel after a failed review submission. 2. Observe if the review appears.	Review does not appear in the admin panel due to a database issue.	Review not displayed due to database issue.	Pass

11. Help center request (Customer)

Test Case ID	Objective	Pre-conditions	Steps to Execute	Expected Results	Actual Results	Status (Pass/Fail)
TC-001	Validate Request Type Selection	Customer selects a valid request type	1. Customer selects a valid request type (e.g., "Technical Issue", "Order Query"). 2. Confirm request type selection.	Valid request type is selected and confirmed.	Valid request type selected, and confirmation displayed.	Pass
TC-002	Validate Invalid Request Type Selection	Customer selects an invalid request type	1. Customer selects an invalid or non-existent request type. 2. Observe system behavior.	Invalid request type is rejected (error message displayed).	Error message displayed for invalid request type selection.	Pass
TC-003	Validate Subject Field Submission	Customer enters a valid subject	1. Customer enters a valid subject (e.g., "Issue with order #1234"). 2. Submit the request.	Subject is valid and request is submitted successfully.	Valid subject entered and request submitted.	Pass
TC-004	Validate Empty or Invalid Subject Field	Customer leaves the subject field empty or enters a non-descriptive subject	1. Customer leaves subject empty or enters a non-descriptive subject	Submission fails due to empty or irrelevant subject (error message displayed).	Error message displayed for empty or irrelevant subject.	Pass

			(e.g., "Help"). 2. Attempt to submit the request.			
TC-005	Validate Description Field Submission	Customer enters a valid description	1. Customer enters a detailed description (e.g., "Product not working as expected"). 2. Submit the request.	Description is valid and request is submitted successfully.	Description entered and request successfully submitted.	Pass
TC-006	Validate Empty or Non-Informative Description Field	Customer enters an empty or non-informative description	1. Customer leaves the description empty or enters something vague (e.g., "Can you help me?"). 2. Attempt to submit the request.	Submission fails due to empty or non-informative description (error message displayed).	Error message displayed for empty or vague description.	Pass
TC-007	Validate Attachments Submission (Valid)	Customer attaches valid files (e.g., .jpg, .png, .pdf)	1. Customer uploads valid attachment(s) (e.g., screenshots, files in supported formats). 2. Submit the request.	Attachments are valid and request is submitted successfully.	Valid attachments uploaded and request submitted.	Pass
TC-008	Validate Attachments	Customer attaches invalid or	1. Customer uploads invalid attachment(	Submission fails due to invalid attachments	Error message displayed for invalid	Pass

	Submission (Invalid)	unsupported files	s) (e.g., .exe, .zip) or no attachment. 2. Attempt to submit the request.	(error message displayed).	attachments.	
TC-009	Validate Submit Request Action	Customer fills all required fields	1. Customer fills all required fields and clicks the "Submit Request" button. 2. Confirm successful submission.	Request is successfully submitted.	Request submitted successfully with all fields filled.	Pass
TC-010	Validate Submit Request Action with Missing Fields	Customer submits the request with missing required fields	1. Customer tries to submit the request without filling all required fields (e.g., subject or description missing). 2. Observe system behavior.	Request submission fails due to missing required fields (error message displayed).	Error message displayed for missing fields.	Pass
TC-011	Validate Request Validation	System validates all fields	1. System checks for empty fields or invalid data (subject, description, attachments). 2. Observe any	All fields validated successfully (no errors).	Fields validated with no errors.	Pass

			validation issues.			
TC-012	Validate Request Confirmation	Request is successfully submitted	1. Customer submits the request. 2. Verify that confirmation email is sent and a success message is displayed.	Confirmation email received and success message displayed ("Your request has been submitted").	Confirmation email received and success message displayed.	Pass
TC-013	Validate Request Submission Failure	System/database error prevents submission	1. Customer submits the request, but system fails due to an error (e.g., database issue). 2. Observe the failure behavior.	Request submission fails (error message displayed).	Error message displayed due to submission failure.	Pass
TC-014	Validate Database Entry for Request	Request is successfully submitted	1. Customer submits the request. 2. Verify that the request is saved in the database.	Request is successfully saved in the database.	Request successfully saved in the database.	Pass
TC-015	Validate Request Display in Admin Dashboard	Request is saved in the database	1. Admin accesses the admin dashboard. 2. Check if the request appears with all details.	Request appears in the admin dashboard with all details.	Request displayed correctly in admin dashboard.	Pass

TC-016	Validate Request Not Displayed in Admin Dashboard	Database/system error prevents display	1. Admin accesses the admin dashboard. 2. Observe if the request appears in the dashboard after failure.	Request does not appear in the admin dashboard due to a system/database issue.	Request not displayed due to database issue.	Pass
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12. Send complaint (Staff)

Test Case ID	Objective	Pre-conditions	Steps to Execute	Expected Results	Actual Results	Status (Pass/Fail)
TC-001	Validate Description Field Submission	Staff enters a valid description	1. Staff enters a detailed description (e.g., "Order communication error"). 2. Submit the complaint.	Description is valid and complaint is submitted successfully.	Description "Order communication error" entered.	Pass
TC-002	Validate Empty or Vague Description Field	Staff enters an empty or vague description	1. Staff leaves description empty or enters something vague (e.g., "There is an issue"). 2. Attempt to submit the complaint.	Submission fails due to empty or vague description (error message displayed).	Error message displayed: "Description cannot be vague."	Pass
TC-003	Validate Submit Complaint Action	Staff fills all required fields	1. Staff fills all required fields and clicks "Submit Complaint". 2. Confirm	Complaint is successfully submitted.	Complaint successfully submitted.	Pass

			successful submission.			
TC-004	Validate Submit Complaint Action with Missing Fields	Staff submits complaint with missing required fields	1. Staff submits the complaint without filling all required fields (e.g., missing subject or description). 2. Observe system behavior.	Submission fails due to missing required fields (error message displayed).	Error message displayed: "Required fields missing."	Pass
TC-005	Validate Complaint Validation	System validates all fields	1. System checks for empty fields or invalid data (subject, description). 2. Observe any validation issues.	All fields validated successfully (no errors).	No validation errors encountered.	Pass
TC-006	Validate Complaint Submission Confirmation	Complaint is successfully submitted	1. Staff submits the complaint. 2. Verify that a confirmation message is displayed.	Confirmation message displayed ("Complaint submitted successfully").	Confirmation message displayed: "Complaint submitted."	Pass
TC-007	Validate Complaint Submission Failure	System/database error prevents submission	1. Staff submits the complaint, but system fails due to an error (e.g., database issue). 2. Observe failure.	Complaint submission fails (error message displayed).	Error message displayed: "System error, try again later."	Pass

TC-008	Validate Database Entry for Complaint	Complaint is successfully submitted	1. Staff submits the complaint. 2. Verify that the complaint is saved in the database.	Complaint is successfully saved in the database with a unique ID.	Complaint saved with ID #12345.	Pass
TC-009	Validate Complaint Display in Admin Dashboard	Complaint is stored in the database	1. Admin accesses the Complaint Management section. 2. Check if the complaint appears with all details.	Complaint appears in the admin dashboard with all details.	Complaint displayed in dashboard with all details.	Pass
TC-010	Validate Complaint Not Displayed in Admin Dashboard	Failure in database update or system issue prevents display	1. Admin accesses the admin dashboard. 2. Observe if the complaint appears in the dashboard.	Complaint does not appear in the admin dashboard due to failure in database update.	Complaint not displayed due to database failure.	Pass